

Mustafa Maher Zalam

Third-Year Electronic Engineering Student at Egyptian E-Learning University (EELU), DEPI trainee, and interned at CodeVita.

Specializing in data analytics, machine learning, business intelligence, automation, and clean problem-solving. I am passionate about transforming raw data into actionable insights, building automated pipelines, and developing robust engineering solutions that drive informed decision-making.

CERTIFICATIONS

Google Data Analytics
DEPI AI & Data Science

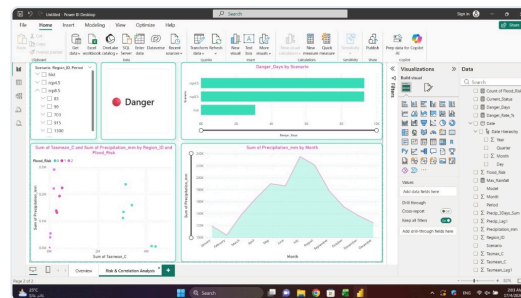
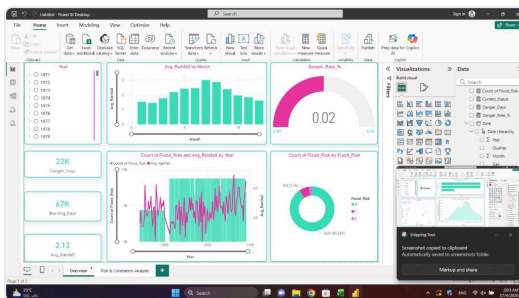


Flood Risk Assessment Dashboard

INTERACTIVE BUSINESS INTELLIGENCE SOLUTION

A comprehensive Power BI geospatial heatmap dashboard developed for real-time flood risk assessment. It systematically processes environmental sensor data alongside historical flood patterns to generate dynamic risk heatmaps across geographic regions.

- Monitors water levels and precipitation data through interactive visualizations.
- Features rigorous time-series analysis and automated anomaly detection alerts.
- Enables comparative scenario modeling essential for emergency response planning.

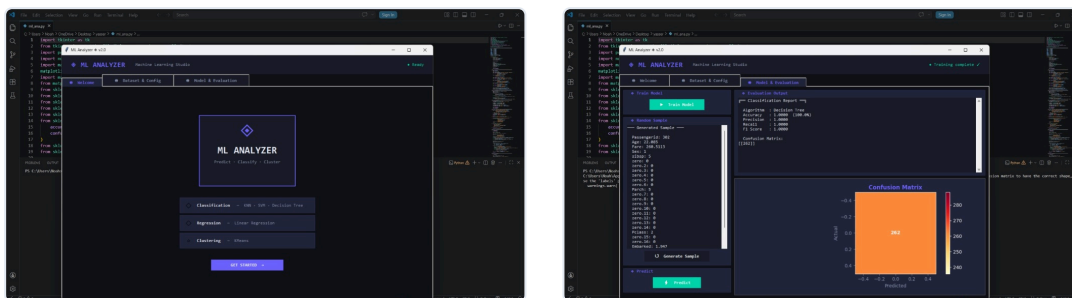


Machine Learning Analyzer

AUTOMATED ML MODEL EVALUATION SYSTEM

A sophisticated Python-based application featuring a dark-themed graphical user interface for automated machine learning model evaluation. It streamlines testing workflows by dynamically generating comprehensive evaluation metrics and performance analysis reports.

- Automates the generation and visualization of confusion matrices.
- Provides robust multi-model comparison with detailed accuracy metrics.
- Offers built-in classification report automation and exportable analysis capabilities.

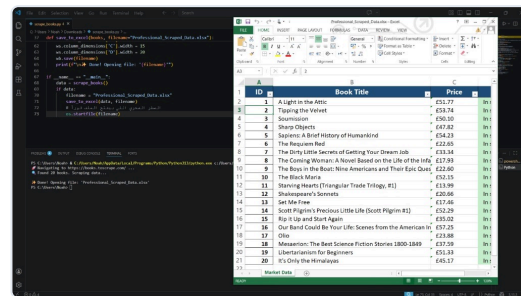
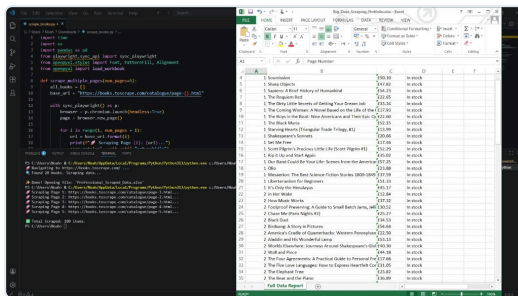


Web Scraping Automation

HIGH-EFFICIENCY DATA EXTRACTION PIPELINE

An engineered Python-based web scraping solution utilizing Selenium for robust and automated data extraction from dynamic websites. It reliably navigates asynchronously loading pages to compile accurate datasets.

- Implements intelligent waiting strategies and comprehensive error handling.
- Incorporates data validation, rate limiting, and automatic retry logic for network stability.
- Exports structured, clean data directly to Excel files for immediate analytical use.

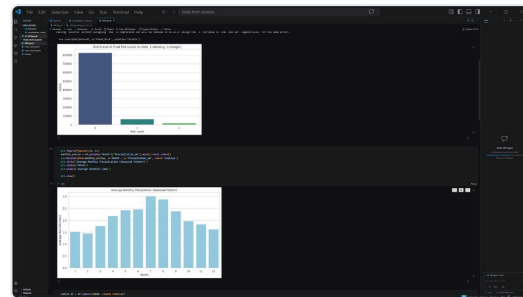
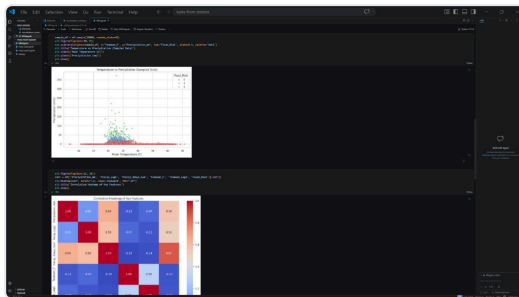


Advanced Data Visualization

COMPREHENSIVE DATA STORYTELLING

A series of comprehensive data storytelling dashboards crafted using modern BI tools. They transform complex, raw datasets into intuitive, actionable business insights through strategic layout and visual hierarchy.

- Strategically selects appropriate chart types based on data dimensions.
- Incorporates interactive elements such as responsive filters and parameter controls.
- Applies robust data aggregation techniques to optimize query execution and loading times.



Technical Skills Summary

CORE COMPETENCIES

PYTHON

Pandas, NumPy, Matplotlib, Seaborn for deep data manipulation, exploratory analysis, and bespoke visualization.

DATA ANALYTICS

Statistical analysis, methodology formulation, data cleaning, processing pipelines, and data engineering fundamentals.

BI & DASHBOARDS

Power BI, Tableau, DAX calculations, interactive data modeling, and end-to-end business intelligence solutions.

MACHINE LEARNING

Scikit-learn, model development, cross-validation metrics, automated predictive analytics, and benchmarking.

WEB SCRAPING

Selenium, Playwright, automated browser navigation, asynchronous loading capture, and error handling structures.

SQL / DATA HANDLING

Query optimization, relational database design, rapid data extraction, and managing large volume datasets effectively.

Get in Touch

LET'S COLLABORATE

I am actively open to internships and professional opportunities in data analytics, business intelligence, machine learning, and data automation. Let's connect to discuss how my engineering background and analytical skill set can contribute to impactful data-driven projects.

